



THE NATIONAL COLLEGE
AUTONOMOUS
BASAVANAGUDI, BANGALORE – 560 004

DEPARTMENT OF
MATHEMATICS

PROPOSED, SCHEME AND
SYLLABUS

For the BACHELOR OF SCIENCE
(B.Sc.),

Mathematics Papers of I and II Semesters
And I Semester BCA

Syllabus for B.Sc. with Mathematics as one of the Major Subjects

SEMESTER – I

Theory	Algebra-I Calculus-I Geometry(DCMT1)
Teaching Hours : 04 Hours/Week	Credits: 03
Duration of Exam: 03 Hours	Maximum Marks:100 (Exam 80 + IA 20)

Course Learning Outcomes:

The overall expectation from this course is that the student builds a basic understanding on Algebra, Calculus and Geometry. The broader course outcomes are listed as follows. At the end of this course, the student will be able to:

- solve the system of homogeneous and non-homogeneous linear system of 'm' equations in 'n' variables by using concept of rank of matrix.
- find eigenvalues and eigenvectors.
- find n^{th} derivatives of some standard functions.
- solve problems on partial differentiation, Jacobians and related properties.
- find the reduction formulae and apply Leibnitz Rule.
- identify geometrical aspects of planes and sphere in 3D.

Algebra-I

Unit I Matrices

14 Hours

Recapitulation of matrices, Elementary row and column transformations (operations). equivalent matrices, theorems on it. Row reduced echelon form. Normal forms of a matrix, Rank of a matrix, problems. Homogenous and non-homogenous systems of linear equations in unknowns, Consistency Criterion - Criterion for uniqueness of solutions. Eigenvalues and Eigenvectors of a square matrix of order 2 and 3, standard properties. Cayley-Hamilton theorem with proof, Finding A^{-1} , A^{-2} and A^2 , A^3 , A^4 .

Calculus -I

Unit II Differential Calculus

14 Hours

Successive differentiation: An n^{th} derivative of the function e^{ax+b} , $(ax+b)^n$, $\log(ax+b)$, $\sin(ax+b)$, $\cos(ax+b)$ $e^{ax} \sin(bx+c)$, $e^{ax} \cos(bx+c)$ and problems, Leibnitz theorem with proof and its applications. Partial differentiation- function of two and three variables- first and higher order derivatives. Homogeneous function- Euler's theorem and its extension with proof, Total derivative and differentiation of implicit function and composite function problems Jacobian properties and problems.

Integral Calculus

Unit III Integral Calculus

14 Hours

Integral Calculus Recapitulation of integration reduction formulas for $\int \sin^n x dx$, $\int \cos^n x dx$, $\int \sin^m x \cdot \cos^n x dx$, $\int \sec^n x dx$, $\int \operatorname{cosec}^n x dx$, $\int \tan^n x dx$, with definite limit problem differentiation under integral sign Leibnitz rule problems.

Analytical Geometry

Unit IV Analytical Geometry

14 Hours

Analytical geometry of three-dimensional, recapitulations of elements of three-dimensional geometry. Planes (intercept form, normal form and passing through three points with proof) distance from a point to a plane, angle between two planes, bisectors of angle between two planes and problems, standard equation of sphere and orthogonal spheres, standard equation of right circular cone and right circular cylinder and problems.

Reference Books:

1. University Algebra -N.S. Gopala Krishnan, New Age International (P) Limited
2. Theory of Matrices - B S Vatsa, New Age International Publishers.
3. Matrices - A R Vasista, Krishna Prakashana Mandir, 2013
4. Differential Calculus - Shanti Narayan, S. Chand & Company, New Delhi.
5. Applications of Calculus, Debasish Sengupta, Books and Allied (P) Ltd., 2019.
6. Calculus – Lipman Bers, Holt, Rinehart & Winston.
7. Calculus - S Narayanan & T. K. Manicavachogam Pillay, S. Viswanathan Pvt. Ltd., vol. I & II.
8. Schaum's Outline of Calculus - Frank Ayres and Elliott Mendelson, 5th edition USA: Mc.Graw
9. Differential Calculus - N p Bali, India: Laxmi Publications (P) Ltd., 2019.
10. Introduction to Calculus and Analytical Geometry - G. B. Thomas and R. L. Finney, Narosa Publishing House, 2010
11. Differential Geometry of Three Dimension - C. E. Weatherburn, Khosla Publishing House, 2020.
12. An elementary treatise on the differential calculus: with applications and numerous examples - J Edwards, B. g print. Charleston, USA: BiblioBazaar, 2010.
13. Calculus - H. Anton, I Birens and S. Davis, John Wiley and Sons, Inc, 2002.

Web links:

1. [http://www.cs.columbib.edu/~zeph/32\\$0allectures.html](http://www.cs.columbib.edu/~zeph/32$0allectures.html)
2. <http://www.themathPage.com>
3. <http://ocw.mit.edu/courses/mathematics>
4. <http://planetmath.org/encyclopedia/TopicsOnCalculus.html>
5. <http://ocw.mit.edu/OcwWeb/Mathematics/18-01Fall-2005/CourseHome/index.html>
6. <http://mathworld.wolfram.com/Calculus.html>
7. <http://ocw.mit.edu/courses/mathematics>
8. <http://www.univie.ac.at/future/media/more/galerie.html>
9. <http://mathworld.wolfram.com/AnalyticGeometry.html>
10. <http://www.nptelvideos.in/2012/11/mathematics.html>
11. <https://www.my-mooc.com/en/categorie/mathematics>

12. www.python.org
13. www.rosettacode.org
14. <http://faculty.msmary.edu/heinold/python.html>
15. <https://kitchingroup.cheme.cmu.edu/pycse/pycse.html>

Practical	Algebra-I Calculus-I Geometry(DCMP1)	
Teaching Hours : 04 Hours/Week		Credits: 02
Duration of Exam: 03 Hours		Maximum Marks: 50 (Exam 40 + IA 10)

Course Learning Outcomes: This course will enable the students to Learn, fundamentals and implement the python programming language, and will enable to:

1. learn fundamentals of python
2. solve problem on algebra and calculus
3. acquire knowledge of applications of algebra and calculus
4. solve problems related to analytical geometry

Practical/lab work to be performed in computer lab using Python

Suggested Programs

1. Introduction to Python
2. Basics of software with simple examples.
3. Basics of software with simple examples.
 - a. compare two numbers using if statements
 - b. sum of natural numbers using while loop
 - c. finding the factors of a number using for loop
 - d. to check the given number is prime or not (use if.....else statement)
 - e. find the factorial of a number(use... if...if...else)
 - f. simple programmes to illustrate logical operators (and or not)
4. Computation of a rank of matrix by row reduced and normal forms
5. Solving the system of homogeneous and non-homogeneous linear equations
6. Computation of inverse of a matrix by using Cayley-Hamilton theorem
7. Finding the nth derivative of a function without Leibnitz theorem
8. Finding the nth derivative of a function with Leibnitz theorem
9. Partial differentiation of some standard functions.
10. Verification of Eulers theorems with examples
11. Program to find Jacobians
12. Program to find reduction formula with or without limits
13. Program to find angle between the two planes
14. Program to find equation and plot sphere
15. Program to find equation and plot cones
16. Program to find equation and plot cylinders

Note: To implement the above programs, there should be not more than 15 students per batch.

Theory	DISCRETE STRUCTURE (24BCA11)	
Teaching Hours : 03 Hours/Week		Credits: 03
Duration of Exam: 03 Hours		Maximum Marks:100 (Exam 80 + IA 20)

Syllabus for I semester BCA

Course Outcome:

- 1 Apply Set theory, Relations, Functions and Mathematical Induction for solving problems
- 2 Getting familiar with counting techniques and Mathematical Logic, and apply the same to address the real life problems
- 3 Learn different operations of a Matrix and apply them to solve real life problems.
- 4 Apply graph theory concepts to solve the problems of real world.

UNIT – I

8 Hours

Set Theory: Fundamentals of Set theory, Set Operations, Laws of Set Theory, Counting and Venn Diagrams, Cartesian Product, Relations, Types of Relations, Functions, Types of Functions, Function Composition, Inverse Functions.

UNIT – II

11 Hours

Logic and Counting: Fundamentals of Logic, Propositional Logic, Logical Connectives and Truth Tables, Logic Equivalence, Tautology and Contradiction. Basics of counting, Counting Principles, Pigeonhole Principle, Permutation, Combinations.

UNIT - III

11 Hours

Matrices: Basics of Matrix, Types of Matrices, Operations on Matrices, Inverse of a matrix, Solution for system of linear equations, Determinant, Properties of Determinant, Cramer's Rule, Introduction to Eigen Values and Eigen Vectors.

UNIT - IV

12 Hours

Graph Theory: Graphs: Introduction, Representing Graphs, Operations on graphs, Directed Graphs Graph Isomorphism, Paths, Cycles, Euler Graph, Hamilton Graph, Planar Graphs. Trees: Introduction, Applications of Trees, Spanning Trees, Minimum Spanning Trees, Prim's and Kruskal's Algorithms.

Text Book :

- 1 Ralph P. Grimaldi: Discrete and Combinatorial Mathematics, 5th Edition, Pearson Education.
- 2 Richard Bronson, Schaum's Outline of Matrix Operations, McGraw-Hill publications, 2nd Edition,

Reference Books :

- 1 Gregory Hartman, Fundamentals of Matrix Algebra, Third Edition
- 2 Gary Haggard, John Schlipf, Discrete Mathematics for Computer Science, , Thomson Books

SEMESTER II

Theory	Algebra-II, Calculus-II & Polar Coordinates(DCMT2)	
Teaching Hours : 4 Hours/Week		Credits: 03
Duration of Exam: 03 Hours		Maximum Marks: 100 (Exam 80 + IA 20)

Course Learning Outcomes:

The overall expectation from this course is that the student builds a basic understanding on Algebra, Calculus and Polar Coordinates. The broader course outcomes are listed as follows. At the end of this course, the student will be able to understand:

- the mathematical structure called Groups.
- the fundamental concepts of limits, continuity and differentiability
- identify and apply the intermediate value theorem and L' Hospital's rule.
- interpret the geometric and physical meaning of derivatives.
- problems related to angle between radius vector and tangent, angle between two curves.
- express the curves in pedal form, derivative of an arc in different forms.
- center of curvature, asymptotes, evolutes and envelopes of the given curve.
- length of an arc, area of plane curves and surface area, volume of revolution.

Algebra II

Unit I Groups

14 Hours

Binary operations algebraic structure- problems on finding identity and inverse definition of semi group and group, abelian group, problems on finite and infinite groups, properties of groups with proof- standard problems on groups-finite semi group with both the cancellation laws in a group- any group of orders less than five is abelian, subgroups theorems on subgroups with proof problems.

Calculus II

Unit II Differential Calculus

14 Hours

Limits and continuity, differentiability and properties of continuous functions. Intermediate value theorem, mean value theorem, Rolle 's Theorem, Lagrange's mean value theorem, Cauchy mean value theorem and examples. Indeterminate forms and evaluation of limits using L'Hospital rule.

Polar Coordinates

Unit III Polar Coordinates

14 Hours

Polar coordinates, angle between the radius vector and tangent, angle of intersection of two curves (polar forms), length of perpendicular from pole to the tangent, pedal equation, derivative of an arc in Cartesian parametric and polar forms, curvature of plane curve, radius of curvature formula in Cartesian, parametric polar and pedal forms, center of curvature evolutes, asymptotes, singular points and double points.

Integral calculus

Unit IV Integral calculus

14 Hours

Application of integral calculus computation of length of arc, plane area and surface area and volume of solids of evaluations for standard curves in Cartesian and polar forms.

Improper integrals of first, second and third kinds with examples, improper integral has the limit of proper integral.

Reference Books:

1. A Course in Abstract Algebra, Vijay K Khanna and S K Bhambri, Vikas Publications. 5 Edition, 2022
2. Elements of Number Theory; I. M. Vinogradov. Courier Dover Publications, 2016
3. Differential Calculus, Shanti Narayan, S. Chand & Company, New Delhi. 2005
4. Integral Calculus, Shanti Narayan and PK Mittal, S. Chand and Co. Pvt. Ltd., 2005
5. Schaum's Outline of Calculus, Frank Ayres and Elliott Mendleson, The McGraw Hill Companies, 2013
6. Mathematical Analysis, S C Malik, Wiley Eastern, 6 Edition, 2017.
7. Differential Calculus, New ed. New Delhi,-N P Bali, India: Laxmi Publications (P) Ltd., 2019
8. An elementary treatise on the differential calculus: with applications and numerous example-J Edwards, Reprint. Charleston, USA: BiblioBazaar, 2010.
9. Algebra, 2nd ed. -Michael Artin, New Delhi, India: PHI Learning Pvt. Ltd., 2011.
10. A First Course in Modern Algebra, 11th ed-Vashista: Krishna PrakasanMandir, 2023.
11. John B Fraleigh, A First course in Abstract Algebra, 7th ed.: Narosa Publishing House., 2003
12. R Balakrishnan and N.Ramabadran, A Textbook of Modern Algebra, 1st ed. New Delhi, India: Vikas publishing house pvt. Ltd., 1991

Web Links

1. <http://www.themathPage>
2. <http://ocw.mit.edu/courses/mathematics/>
3. <http://planetmath.org/encyclopedia/TopicsOnCalculus.html>
4. <http://mathworld.wolfram.com/Calculus.html>
5. <http://ocw.mit.edu/courses/mathematics/>
6. <http://www.univie.ac.at/future.media/moreIgalerie>
7. <http://www.nptelvideos.in/2012/77/mathematics.html>
8. <https://www.my-mooc.com/en/categories/mathematics>
9. www.python.org
10. www.rosettacode.org
11. <http://faculty.msmar.edu/heinold/python.html>
12. <https://litchingroup.chem.combed/physe/physe.html>

Practical	Algebra-II, Calculus-II and Polar Coordinate(DCMP2)
Teaching Hours : 04 Hours/Week	Credits: 02
Duration of Exam: 03 Hours	Maximum Marks: 50 (Exam 40 + IA 10)

Course Learning Outcomes: This course will enable the students to write python code to:

1. solve problems on algebra and calculus.
2. acquire knowledge of applications of algebra and calculus
3. plot the curves in different forms
4. find surface area and volume of revolution

Practical/lab work to be performed in computer lab using Python

Suggested Programs

1. Verifying whether given operator is binary or not.
2. To find identity and inverse element of a group
3. Program to construct Cayley's table for a given set.
4. Program to illustrate continuity of a function
5. Program to illustrate differentiability of a function
6. Program to verify Rolle's Theorem
7. Program to verify Lagrange's theorem
8. Evaluation of limits by L'Hospital rules.
9. Finding the angle between the radius vector and tangent
10. Finding the angle between two curves
11. Finding the radius of curvature of the given curve
12. Plotting of standard Cartesian curves
13. Plotting of standard polar curves
14. Plotting of standard parametric curves
15. Program to compute surface area of revolution.
16. Program to compute volume of revolution.

Note: To implement the above programs, there should be not more than 15 students per batch.

Semester	Course Code	Paper Title	Teaching Hours / Week	Marks		Duration of Exam in Hours	Credits
				Exam	IA		
I	Theory	Algebra-I, Calculus-I & Geometry	04	80	20	03	03
	Practical	Algebra-I, Calculus-I & Geometry	04	40	10	03	02
I	Theory	Discrete Structure	03	80	20	03	03
II	Theory	Algebra-II, Calculus-II & Polar Coordinates	04	80	20	03	03
	Practical	Algebra II, Calculus II & Polar Coordinates	04	40	10	03	02

Format of question paper I semester B.Sc.

Question No.	Topic and No. of subdivisions to be set in the topic		No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
I	Algebra-I	3	10	2	20
	Differential Calculus	3			
	Integral Calculus	3			
	Geometry	3			
II	Algebra-I	5	3	5	15
III	Differential Calculus	5	3	5	15
IV	Integral Calculus	5	3	5	15
V	Geometry	5	3	5	15
				Total	80

Practical Question paper format

PART-I 1 question out of 2 questions	1*8=8 marks
PART-II 1 question out of 2 questions	1*8=8 marks
PART-III 1 question out of 2 questions	1*8=8 marks
PART-IV 1 question out of 2 questions	1*8=8 marks
Record	5 marks
Viva	3 marks
Total	40 marks

Format of question paper I semester BCA.

Question No.	Topic and No. of subdivisions to be set in the topic		No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
I	Set theory	3	10	2	20
	Logic and counting	3			
	Matrices	3			
	Graph theory	3			
II	Set theory	5	3	5	15
III	Logic and counting	5	3	5	15
IV	Matrices	5	3	5	15
V	Graph theory	5	3	5	15
				Total	80

Format of question paper II semester B.Sc.

Question No.	Topic and No. of subdivisions to be set in the topic		No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
I	Algebra-II	3	10	2	20
	Differential Calculus-II	3			
	Polar Coordinates	3			
	Integral Calculus	3			
II	Algebra-II	5	3	5	15
III	Differential Calculus-II	5	3	5	15
IV	Polar Coordinates	5	3	5	15
V	Integral Calculus	5	3	5	15
				Total	80

Practical Question paper format

PART-I 1 question out of 2 questions	1*8=8 marks
PART-II 1 question out of 2 questions	1*8=8 marks
PART-III 1 question out of 2 questions	1*8=8 marks
PART-IV 1 question out of 2 questions	1*8=8 marks
Record	5 marks
Viva	3 marks
Total	40 marks



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DEPARTMENT OF MATHEMATICS

CHOICE BASED CREDIT SYSTEM

Academic year
2025-2026

III and IV SEMESTER (SEP)

Name of the Degree Program

:Bachelor of Science-BSc

Discipline Course

:Mathematics

Starting Year of Implementation

**:2024-25 (I & II Semesters)
2025-26 (III & IV Semesters)
2026-27 (V & VI Semesters)**

Programme Outcomes(PO): By the end of the program the students will be able to achieve:

PO1	Disciplinary Knowledge: Bachelor degree with Mathematics as one of subjects in chosen combination is the culmination of in-depth knowledge of Algebra, Calculus, Geometry, differential equations and several other branches of pure and applied mathematics. This also leads to study the related areas such as computer science and other allied subjects.
PO2	Communication Skills: Ability to communicate various mathematical concepts effectively using examples and their geometrical visualization. The skills and knowledge gained in this program will lead to the proficiency in analytical reasoning which can be used for modeling and solving of real life problems.
PO3	Critical thinking and analytical reasoning: The students undergoing this programme acquire ability of critical thinking and logical reasoning and capability of recognizing and distinguishing the various aspects of real life problems.
PO4	Problem Solving: The Mathematical knowledge gained by the students through this programme develop an ability to analyze the problems, identify and define appropriate computing requirements for its solutions. This programme enhances students overall development and also equip them with mathematical modeling ability, problem solving skills.
PO5	Research related skills: The completing this programme develop the capability of inquiring about appropriate questions relating to the Mathematical concepts in different areas of Mathematics.
PO6	Information/digital Literacy: The completion of this programme will enable the learner to use appropriate software to solve system of algebraic equation and differential equations.
PO7	Self-directed learning: The student completing this program will develop ability of working independently and to make an in-depth study of various notions of Mathematics.
PO8	Moral and ethical awareness/reasoning: : The student completing this program will develop an ability to identify unethical behavior such as fabrication, falsification or misinterpretation of data and adopting objectives, unbiased and truthful actions in all aspects of life in general and Mathematical studies in particular.

PO9	Lifelong learning: This programme provides self-directed learning and lifelong learning skills. This programme helps the learner to think independently and develop algorithms and Computational skills for solving real word problems.
PO10	Ability to use advanced studies and research in pure and applied Mathematical sciences.
PO11	Students will be able to formulate LPP and solve problem. Students will be able to build and solve transportation and assignment models.
PO12	Enable students to understand and apply foundational concepts, construct valid arguments and potentially pursue advanced studies or research in related fields. Understanding of axioms and their role in mathematical reasoning. Students should be familiar with Boolean algebra, propositional calculus and their application in digital circuits design and other fields

ASSESSMENT

Weightage for the Assessments (in percentage)

Type of Course	Formative Assessment/ I.A.	Summative Assessment (S.A.)
Theory	20%	80%
Practical	20%	80%

Syllabus for B.Sc. with Mathematics as one of the Major Subjects

SEMESTER–III

Theory	BSC3-MT-T1-Algebra-III, Real Analysis-I&Differential Equations-I	
TeachingHours:04Hours/Week		Credits:03
DurationofExam:03 Hours		MaximumMarks:100(Exam80+IA20)

Course Learning Outcomes:

The overall expectation from this course is that the student builds a basic understanding on Algebra, Analysis and Differential Equations . The broader course outcomes are listed as follows. At the end of this course, the student will be able to achieve:

- Link the fundamental concepts of groups and symmetries of geometrical objects.
- Explain the significance of the notions of cosets
- Understand the fundamental properties of the real numbers that lead to define sequence and series, the formal development of real analysis.
- Learn the concept of Convergence and Divergence of a sequence.
- Obtain the skills of analyzing the concepts and applying appropriate methods for testing convergence of a sequence/ series
- Apply the ratio, root, alternating series, and limit comparison tests for convergence and absolute convergence of an infinite series
- Solve first-order non-linear differential equations and linear differential equations.
- To model problems in nature using Ordinary Differential Equations.
- Formulate differential equations for various mathematical models.

ALGEBRA- III

Unit I: Groups-I

(14 lecture hours)

Order of an element of a group – properties and theorems, subgroup generated by an element of a group – coset decomposition of a group, Cyclic groups, properties- modulo relation- index of a group – Lagrange's theorem- consequences, Fermat's theorem and Euler's ϕ function.

REAL ANALYSIS–I

Unit II : Sequences

(14 lecture hours)

Sequences of real numbers, Bounded sequences– properties and theorems, Limit of a sequence, Convergent, divergent and oscillatory sequences– properties and theorems, Algebra of the limits of the sequences, Monotonic sequences– problems and theorems, Behaviour of standard sequences- $\{x^{1/n}\}$, $\{n^{1/n}\}$, $\{(1+\frac{1}{n})^n\}$, $\{x^n\}$, Cauchy's general principle for convergence of a sequence.

Unit III : Infinite Series**(14 lecture hours)**

Definition of convergent, divergent and oscillatory series. Series of nonnegative terms, Geometric series, p-series (Harmonic series). Comparison tests for positive term series: D'Alembert's ratio test, Raabe's test, Cauchy's Root test. Alternating series: Leibnitz's test. Absolute convergence and conditional convergence of a series. Summation of series: Binomial, exponential and logarithmic.

DIFFERENTIAL EQUATIONS-I:**Unit IV: Ordinary Differential Equations****(14 lecture hours)**

Recapitulation of differential equations of first order and first degree, Exact Differential equations. Solutions of Differential equations of first order and higher degree: Equations solvable for x, y . Clairaut's equation. Orthogonal trajectories of Cartesian and polar curves.

Reference Books:

1. M.D. Raisinghania, Ordinary Differential Equations & Partial Differential Equations, S. Chand & Company, New Delhi.
2. J. Sinha Roy and S. Padhy: A course of Ordinary and Partial Differential Equation, Kalyani Publishers, New Delhi.
3. D. Murray, Introductory Course in Differential Equations, Orient Longman (India)
4. W.T. Reid, Ordinary Differential Equations, John Wiley, New Delhi.
5. M.L. Khanna, Differential Equations, Jai Prakash Nath & Co. Meerut.
6. S.L. Ross, Differential Equations, 3rd Ed., John Wiley and Sons, 1984.
7. R.G. Bartle and D.R. Sherbert, Introduction to Real Analysis, 3rd Ed., John Wiley and Sons (Asia) Pvt. Ltd., Singapore, 2015.
8. Gerald G. Biledeau, Paul R. Thie, G.E. Keough, An Introduction to Analysis, 2nd Ed., Jones & Bartlett, 2010.
9. S.K. Berberian, A First Course in Real Analysis, Springer Verlag, New York, 1994.
10. T. Apostol, Mathematical Analysis, Narosa Publishing House
11. M.L. Khanna and L.S. Varhiney, Real Analysis by, Jai Prakash Nath & Co. Meerut.
12. Kreyzig, Advanced Engineering Mathematics, John Wiley, New Delhi.
13. A Course in Abstract Algebra, Vijay K. Khanna and S. K. Bhambri, Vikas Publications.

Web links:

1. <http://www.themathpage.com>
2. <http://www.abstractmath.org>
3. <http://ocw.mit.edu/courses/mathematics/>
4. <http://planetmath.org/encyclopedia/TopicsOnCalculus.html>
5. www.python.org
6. www.rosettacode.org
7. <http://www.univie.ac.at/future.media/moe/galeri.html>
8. <http://tutorial.math.lamar.edu/classes/de/de.aspx>
9. <http://www.sosmath.com/diffeq/diffeq.html>

Practical	BSC3-MT-P1-Algebra-III,Real Analysis-I&Differential Equations-I	
TeachingHours:03Hours/Week		Credits:02
DurationofExam:03 Hours		MaximumMarks:50(Exam40+IA10)

Course Learning Outcomes: This course will enable the students to gain hands-on experience of

- Free and Open Source software(FOSS)tools or computer programming.
- Findingcomplementaryfunctionandparticularintegraloflinearandhomogeneous differential equations.
- Acquireknowledgeofapplicationsofrealanalysisanddifferentialequations.
- Verification of convergence/divergence of different types of series

Practicals list

1. Program to find all possible cosets of the given finite group.
2. Program to find generators and corresponding possible subgroups of a cyclic group.
3. Program to verify Lagrange's theorem with suitable examples.
4. Program to verify Euler's ϕ Function for a given finite group.
5. Convergence of sequences.
6. Convergence of geometric series.
7. Convergence of p-series and D'Alembert's Test.
8. Examples on alternating series using Leibnitz's theorem.
9. Summation of exponential and logarithmic series.
10. Summation of binomial series.
11. Verification of exactness of a differential equation.
12. Solutions of differential equations that are solvable for p, x, y.
13. To find the singular solution by using Clairaut's form.

SEMESTER–III

Theory: Elective– I	Linear programming
Teaching Hours:02Hours/Week	Credits:02
Duration of Exam:90minutes	Maximum Marks:50(Exam40+IA10)

Course Learning Outcomes:

1. Apply mathematical and statistical methods to solve real world problems.
2. Develop and implement to optimization models.
3. Communicate complex technical information effectively.

Unit-1

Linear Programming Problems (14lecturehours)

Introduction to LPP-Graphs of linear inequalities, classification of solutions, Formulation of LPP in standard form, solution by Graphical Method, Simplex Method (Maximization with Two and three variables, minimization with two variables)

Unit-2

Transportation problems (14lecturehours)

Mathematical formulation of the Transportation problem(TP), finding initial basic feasible solution of TP using the North-West Corner Rule, Least Cost method, Matrix minima method, Vogel's Approximation Method for finding the optimal solution for TP, and formulation and solutions of the Assignment problem by using the Hungarian method.

References/Suggested Readings:

1. G.Hadley: Linear Programming.Narosa, Reprint,2002.
2. G.Hadley: Linear Algebra, Narosa, Reprint,2002.
3. HamdyA.Taha:Operations Research-An Introduction,PrenticeHall,9thEdition,2010.
4. A.Ravindran,D.T.Phillips and JamesJ.Solberg: Operations Research-Principles and Practice, John Wiley & Sons, 2005.
5. F.S.Hillier.G.J.Lieberman: Introduction to OperationsResearch-ConceptsandCases,9th Edition, Tata Mc-Graw Hill, 2010.

SEMESTER-IV

Theory	BSC4-MT-T2 Algebra-IV, Integral Transforms & Differential Equations-II	
TeachingHours:04Hours/Week		Credits:03
DurationofExam:03 Hours		MaximumMarks:100(Exam80+IA20)

Course Learning Out comes: This course will enable the students to:

- Explain the significance of normal subgroups and factor groups.
- Learn the quotient groups, concepts of homomorphism, isomorphism and properties
- Solve the ordinary differential equations of the first order and second order
- Solve linear and non-linear differential equations using various methods; and apply these methods to solving some physical problems. Evaluate Laplace transforms of certain functions, find Laplace transforms of derivatives and of integrals.
- Understand properties of inverse Laplace transforms, find inverse Laplace transforms of derivatives and of integrals.
- Solve ordinary differential equations with constant/variable coefficients by using Laplace transform method.
- Comprehend the properties of Fourier transforms and solve problems related to finite Fourier transforms.

ALGEBRA- IV

Unit I: Groups-II

(14 lecture hours)

Normal Subgroups, Quotient groups, Homomorphism and Isomorphism of groups, kernel and image of a homomorphism, Normality of the kernel, Fundamental theorem of homomorphism, Properties related to isomorphism, Permutation group, Cayley's Theorem.

DIFFERENTIAL EQUATIONS-II:

Unit II: Linear Differential Equations

(14 lecture hours)

Linear differential equations of n^{th} order with constant coefficients. Particular Integrals when the RHS is of the form e^{ax} , $\sin(ax+b)$, $\cos(ax+b)$, x^n , $e^{ax}V$, where V is a function of x . Cauchy-Euler equations, Solving second order ODE by changing independent variable, by changing dependent variable and Method of variation of parameters.

Integral Transforms

Unit III: Laplace transforms

(18 lecture hours)

Definition, Basic Properties. Laplace transforms of some standard functions, Inverse Laplace transforms, Convolution theorem, Laplace transform of periodic functions. Laplace transform of derivative and integral of a function. Heaviside function. Dirac-delta function. Solutions of differential equations by using Laplace transforms.

Unit IV: Fourier series

(10 lecture hours)

Periodic functions. Fourier Coefficients. Fourier series of functions with period $2L$. Fourier series of even and odd functions. Half range Cosine and Sine series.

Reference Books:

1. D.A. Murray, Introductory Course in Differential Equations, Orient and Longman
2. H.T.H. Piaggio, Elementary Treatise on Differential Equations and their Applications, CBS Publisher & Distributors, Delhi, 1985.
3. G.F. Simmons, Differential Equations, Tata McGraw Hill.
4. S.L. Ross, Differential Equations, 3rd Ed., John Wiley and Sons, India, 2004.
5. M. D. Raisinghania, Ordinary Differential Equations & Partial Differential Equations, S. Chand & Company, New Delhi.
6. R. Murray and L. Spiegel, Laplace Transforms, Schaum's Series
7. Sudhir Kumar, Integral Transform Methods in Science & Engineering, CBS Engineering Series, 2017.
8. Murray R. Spiegel L., Fourier Transforms, Schaum's Series,
9. Earl David Rainville and Philip Edward Bedient—A short course in Differential Equations, Prentice Hall College Div; 6th Edition.
10. Sathya Prakash, Mathematical Physics, S. Chand and Sons, New Delhi.
11. A Course in Abstract Algebra, Vijay K. Khanna and S.K. Bhambri, Vikas Publications

Web links:

1. <http://www.themathpage.com>
2. <http://www.abstractmath.org>
3. <http://ocw.mit.edu/courses/mathematics/>
4. <http://planetmath.org/encyclopedia/TopicsOnCalculus.html>
5. www.python.org
6. www.rosettacode.org
7. <http://www.univie.ac.at/future.media/moe/galeri.html>
8. <http://tutorial.math.lamar.edu/classes/de/de.aspx>
9. <http://www.sosmath.com/diffeq/diffeq.html>

Practical	BSc4-MT-P2Algebra-IV,IntegralTransforms&DifferentialEquations-II
TeachingHours:03Hours/Week	Credits:02
DurationofExam:03 Hours	MaximumMarks:50(Exam40+IA10)

Course Learning Outcomes: This course will enable the students to write python code

- Solve problems on Partial Differential Equations and Integral Forms
- To find Laplace transform of various functions
- To find the Fourier Transform of periodic functions
- To solve differential equations by using Integral transforms.

Programmes:

1. Verification of Normality of a given subgroup
2. Program to verify the given function is homomorphism.
3. Program to verify the given function is isomorphism.
4. Finding complementary function of ordinary differential equations with constant coefficients
5. Finding complementary function and particular integral of ordinary differential equations with constant coefficients .
6. Solution of second order ordinary differential equations with variable coefficients by the method of variation of parameters
7. Finding the Laplace transforms of some standard and periodic functions.
8. Finding the inverse Laplace transform of simple functions.
9. Solution of ordinary linear differential equation using Laplace transform.
10. To find Fourier series of some simple functions with period $2L$
11. To find half range sine and cosine series of some simple functions
12. To find cosine and sine Fourier transform

SEMESTER–IV

Theory	Elective–II Mathematical logic	
TeachingHours:02Hours/Week		Credits:02
DurationofExam:90minutes		MaximumMarks:50(Exam40+IA10)

Course Learning Outcomes: On completion of this course, students are able to:

- Understand the algebraic concepts of Mathematical logic and Boolean algebra
- Concepts of Mathematical logic
- Application of Mathematical logic in switching networks/circuits.

Unit-I: Mathematical Logic

(14lecturehours)

Propositions, logical connectors, truth tables, tautology, contradiction, contingent statements, negations, inverse, converse and contra positive statements. Open sentences and quantifiers, truth sets connectives involving quantifiers.

Methods of proof: Direct proofs, indirect proofs, Contradiction method, contrapositive method and mathematical induction (explanation with simple examples). Rules of Inference (for quantified statements), Validity of Arguments, Principal disjunctive normal forms and Principal conjunctive normal forms.

Unit-II: Fuzzy Logic

Vocabulary of Fuzzylogic-Booleansets-operators-Fuzzysets- FuzzyQuantifiers- Fuzzysetoperators- Operations on Fuzzy sets – Illustrations – Applications.

(14lecture hours)

Text Books:

1. Discrete Mathematics, KennethRosen, 2001.

Reference Books:

1. J.P.Tremblay and R.P.Manohar: Discrete Mathematical Structures with applications to computer science, McGraw Hill (1975).
2. C.L.Liu: Elements of Discrete Mathematics,TataMcGraw-Hill,2000.

SEMESTER-IV

Practical	SEC-I Mathematical Statistics	
TeachingHours:03Hours/Week		Credits:02
DurationofExam:03 Hours		MaximumMarks:50(Exam40+IA10)

Course Learning Outcomes: This course will enable students to

CO1 Acquire knowledge of statistical methods such as tests of hypothesis correlation and regression.

CO2 Identify types of data and visualization, analysis and interpretation.

CO3 Know about elementary probability and probability models. CO4

Employ suitable test procedures for given data set.

CO5 Use resources available on open source (R programming)

CONTENTS:

1. Introduction to R Language
2. Measures of central tendency-Mean, Median and Mode
3. Measures of Dispersion-I
4. Measures of Dispersion-II
5. Curve fitting-Scatter plot
6. Correlation-I
7. Linear Regression
8. Probability-Bayes's theorem
9. Probability distributions- Binomial distribution
10. Probability distributions-Poisson distribution
11. Probability distributions-Normal distribution

Reference Books:

1. Fundamentals of Mathematical Statistics, S C Gupta, VK Kapoor, Sultan Chand, 2020.
2. Statistical Methods, SP Gupta, Sultan Chand, 2020.
3. Mathematical Statistics, JN Kapur, HC Saxena, S Chand, 2020.
4. Open intro Statistics, Diez, Barr, Cetinkaya-Rundel, Openintro, Inc, 2019.
5. Elementary Statistics, Mario F Triola, Pearson, 2018.

Semester	Course Code	Paper Title	Teaching Hours / Week	Marks		Duration of Exam in Hours	Credits
				Exam	IA		
III	Theory	Algebra-III, Real analysis-I & Differential Equation-I	04	80	20	03	03
	Practical	Algebra-III, Real analysis-I & Differential Equation-I	03	40	10	03	02
III	Elective	Linear Programming	02	40	10	1.5	02
IV	Theory	Algebra-IV, Integral Transforms & Differential Equation-II	04	80	20	03	03
	Practical	Algebra-IV, Integral Transforms & Differential Equation-II	03	40	10	03	02
IV	Elective	Mathematical logic	02	40	10	1.5	02
IV	Practical	Mathematical Statistics	03	40	10	03	02

Question No.	Topic and No. of subdivisions to be set in the topic		No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
I	Groups-I Sequences-I Infinte Series Ordinary Differential Equations	3 3 3 3	10	2	20
II	Groups-I	5	3	5	15
III	Sequences-I	5	3	5	15
IV	Infinte Series	5	3	5	15
V	Ordinary Differential Equations	5	3	5	15
				Total	80

Practical Question paper format

PART-I 1 question out of 2 questions	1*15=15 marks (5 marks: Manual + 7 marks: Program + 3 marks: Execution)
PART-II 1 question out of 2 questions	1*15=15 marks (5 marks: Manual + 7 marks: Program + 3 marks: Execution)
Record	5 marks
Viva	5 marks
Total	40 marks

Format of question paper III semester B.Sc- Elective

Question No.	Topic and No. of subdivisions to be set in the topic	No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
1	Linear Programming Problems Transportation problems	3 3	5 2	10
2	Linear Programming Problems Transportation problems	3 3	4 5	20
3	Linear Programming Problems Transportation problems	1 1	1 10	10
TOTAL				40

Format of question paper IV semester B.Sc.

Question No.	Topic and No. of subdivisions to be set in the topic		No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
I	Groups-II Linear Differential Equations-II Laplace Transform Fourier Series and Fourier Transform	3 3 3 3	10	2	20
II	Groups-II	5	3	5	15
III	Linear Differential Equations-II	5	3	5	15
IV	Laplace Transform	5	3	5	15
V	Fourier Series and Fourier Transform	5	3	5	15
				Total	80

Format of question paper IV semester B.Sc- Elective

Question No.	Topic and No. of subdivisions to be set in the topic		No. of subdivisions to be answered	Marks for each sub division	Max. Marks for the question
1	Mathematical logic Fuzzy logic	3 3	5	2	10
2	Mathematical logic Fuzzy logic	3 3	4	5	20
3	Mathematical logic Fuzzy logic	1 1	1	10	10
TOTAL					40

Practical Question paper format

PART-I 1 question out of 2 questions	1*15=15 marks (5 marks: Manual + 7 marks: Program + 3 marks: Execution)
PART-II 1 question out of 2 questions	1*15=15 marks (5 marks: Manual + 7 marks: Program + 3 marks: Execution)
Record	5 marks
Viva	3 marks
Total	40 marks

Practical Question paper format for Mathematical Statistics

PART-I 1 question out of 2 questions	1*15=15 marks (5 marks: Manual + 7 marks: Program + 3 marks: Execution)
PART-II 1 question out of 2 questions	1*15=15 marks (5 marks: Manual + 7 marks: Program + 3 marks: Execution)
Record	5 marks
Viva	3 marks
Total	40 marks